FLIGHT DELAY, CANCELLATION AND AIRLINE PERFORMANCE ANALYSIS USING PYTHON

1. Introduction

This project explores flight delays, cancellations, airline performance, and time-based patterns. I used Python in Google Colab to clean and analyse the data, then built a two-page Power BI dashboard to present the main findings.

The main purpose was to understand flight performance through data analysis and present the results in a way that is easy to understand.

2. Problem Statement and Objectives

Flight delays and cancellations can affect passengers and airline operations. Analysing flight data can help identify patterns in delays and cancellations and compare airline performance.

The objectives of this project were to:

- Explore flight delay and cancellation patterns.
- Compare average departure and arrival delays across airlines.
- Analyse delay variations across days of the week.
- Examine the relationship between flight distance and arrival delay.
- Create meaningful visualisations using Python.
- Build an interactive Power BI dashboard to present the findings.

3. Dataset and Tools

Dataset: T_ONTIME_REPORTING.csv

Tools and technologies used:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Microsoft Power BI
- CSV files

The dataset was used to examine flight operations, delays, cancellations, airline performance, and other relevant flight information for January 2024.

4. Data Preparation

The original dataset contained **547,271 rows**. After cleaning and preprocessing, **542,700 rows and 34 columns** remained.

A total of **4,571 rows were removed** during preprocessing, and the notebook reported zero duplicate rows remaining. Some missing values remained in certain fields, so results involving those fields should be interpreted carefully.

The cleaned dataset was exported as flight_delay_cleaned_preprocessed.csv and used to create the Power BI dashboard.

5. Analysis Approach

I used descriptive statistics, group-by operations, pivot tables, correlation analysis, and visualisations to explore the dataset.

The analysis covered:

- Departure and arrival delay distributions.
- Average departure and arrival delays by airline.
- Flight cancellation patterns.
- Daily arrival delay trends during January 2024.
- Average arrival delay by day of the week.
- Flight distance versus arrival delay.
- Average arrival delay by distance category.
- Flight-status distribution.
- Correlation between selected numerical variables.

These analyses helped me identify important patterns and compare flight performance across different categories.

6. Key Findings

6.1 Overall Flight Performance

The Power BI dashboard shows an average departure delay of **15.71 minutes** and an average arrival delay of **10.36 minutes**.

These figures provide an overview of delays in the analysed dataset.

6.2 Flight Cancellations

The cleaned dataset contains **15,996 cancelled flights out of 542,700 records**, giving a cancellation rate of **2.95%**.

This indicates that cancellations accounted for a small but important portion of the flight records.

6.3 Airline Performance

Among the airlines visible in the Power BI dashboard, **AA has the highest displayed average arrival delay at 19.18 minutes and average departure delay at 22.71 minutes**.

This highlights differences in average delay performance among the displayed airlines. However, the averages alone do not explain why the delays occurred.

6.4 Delay Patterns by Day of the Week

The Python analysis shows that **Tuesday has the highest overall average arrival delay at 16.26 minutes**, while **Thursday has the lowest at 3.71 minutes**.

This suggests that average arrival delays varied across the days of the week during the analysed period.

6.5 Flight-Status Distribution

The Power BI dashboard displays the following flight-status distribution:

- **On Time/Early:** 58.47%
- **Delayed:** 38.58%
- **Cancelled:** 2.95%

The chart provides a simple overview of the proportion of flights in each status category.

6.6 Flight Distance Analysis

The dashboard shows the following average arrival delays by distance category:

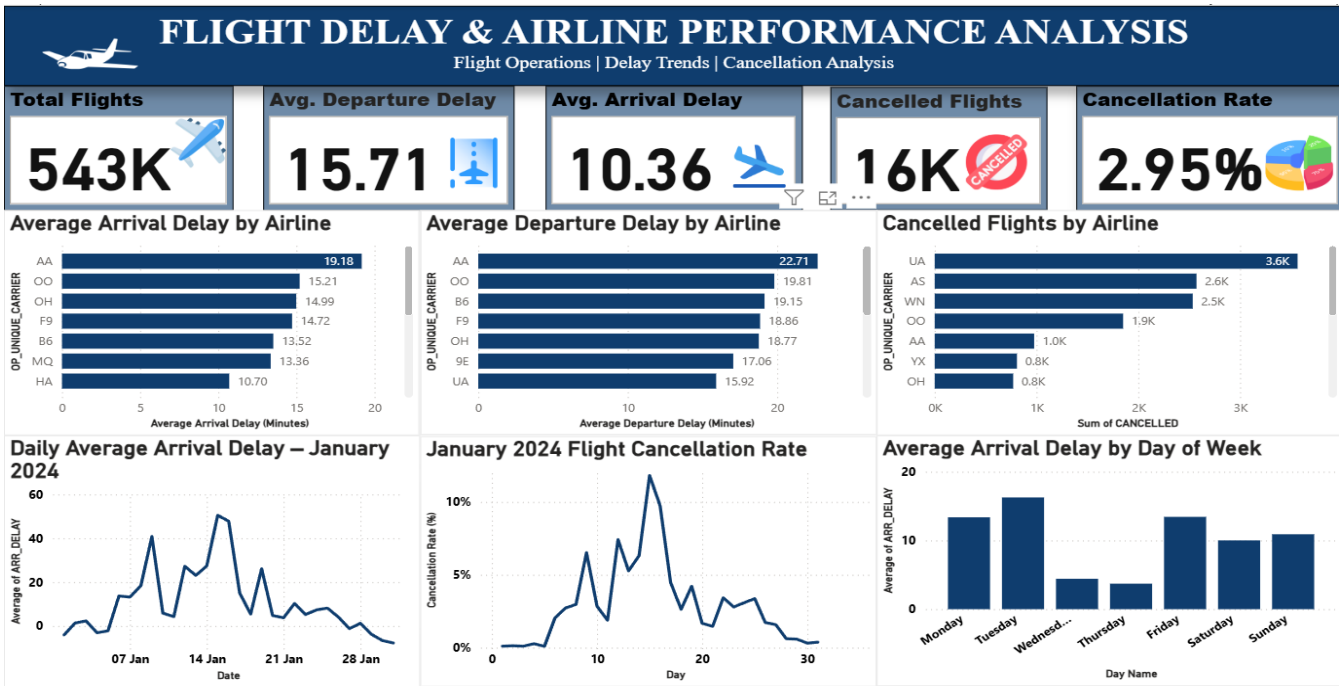
- **Medium-distance flights:** 11.21 minutes
- **Short-distance flights:** 10.51 minutes
- **Long-distance flights:** 6.25 minutes

In this dataset, the long-distance category has the lowest average arrival delay among the three displayed categories. Flight distance alone, however, does not explain all delay patterns.

7. Power BI Dashboard

I created a two-page Power BI dashboard to present the analysis clearly.

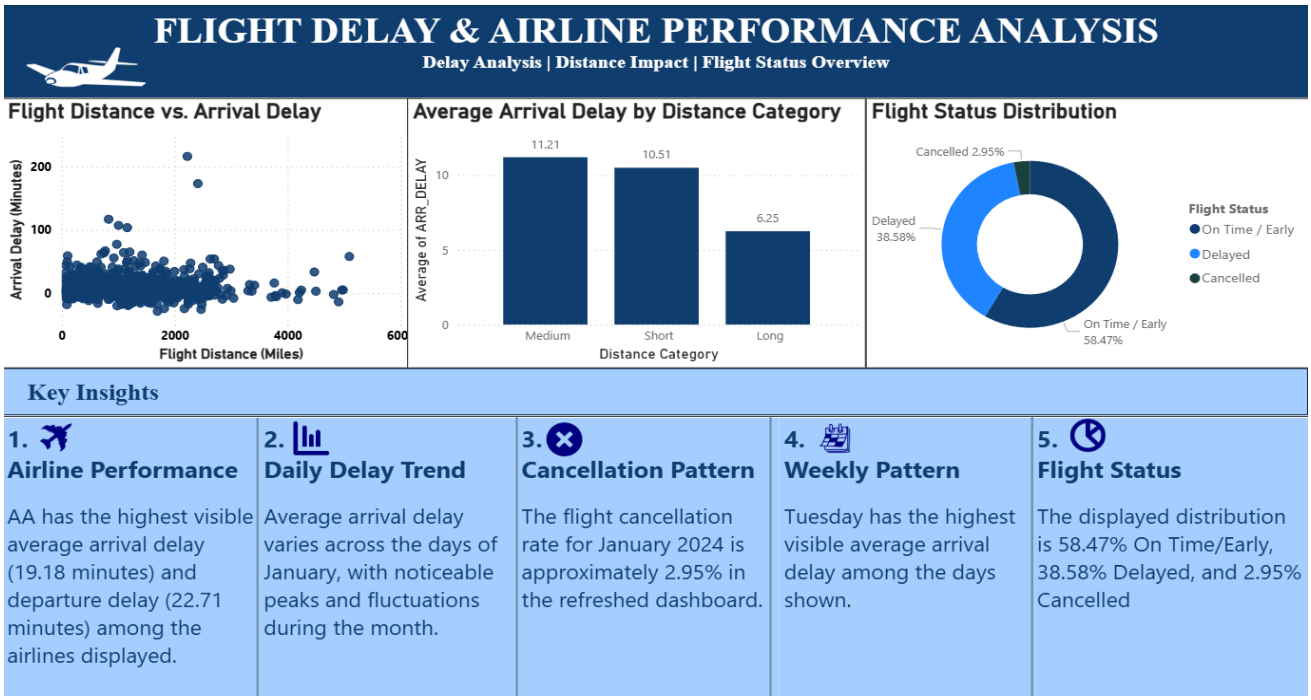
Page 1: Flight Operations Overview



This page includes:

- Total flights.
- Average departure delay.
- Average arrival delay.
- Cancelled flights.
- Cancellation rate.
- Average arrival delay by airline.
- Average departure delay by airline.
- Cancelled flights by airline.
- Daily average arrival delay during January 2024.
- Average arrival delay by day of the week.

Page 2: Delay Analysis and Key Insights



This page includes:

- Flight distance versus arrival delay scatter plot.
- Average arrival delay by distance category.
- Flight-status distribution.
- Key insights summarising airline performance, daily delay trends, cancellations, and weekday patterns.

The dashboard helps present the results in a visual and accessible format.

8. Business Recommendations

Based on the analysis, I recommend the following:

1. **Monitor airline performance:** Review departure and arrival delays regularly to identify patterns that may require further investigation.
2. **Review cancellation patterns:** Examine dates and operating conditions associated with higher cancellation rates.
3. **Study weekday variations:** Compare weekday patterns over longer periods before making scheduling decisions.
4. **Investigate possible causes:** Include additional information, such as weather conditions, airports, routes, and scheduled departure times, to understand the factors associated with delays.
5. **Maintain the dashboard:** Refresh the data regularly and use consistent definitions for delay and cancellation measures.

These recommendations are intended to guide further analysis rather than suggest that the observed patterns prove a specific cause.

9. Limitations

This project analyses flight data for January 2024 only. The findings may not represent patterns across the entire year.

Some missing values remain in the dataset. Airline averages may also be influenced by differences in routes, flight volumes, and operating conditions.

The analysis identifies patterns and relationships but does not establish the causes of delays or cancellations. Additional data and a longer observation period would be useful for a more comprehensive investigation.

10. Future Enhancements

This project can be improved further in the following ways:

- **Larger Dataset:** Use a larger and more recent flight dataset to identify additional patterns and improve the analysis.
- **Automation:** Automate data cleaning and analysis to reduce manual work and improve efficiency.
- **Interactive Dashboards:** Enhance the Power BI dashboard with additional filters and interactive visualizations to explore flight performance more effectively.
- **Predictive Analysis:** Apply machine learning techniques to predict flight delays and identify flights that may have a higher risk of cancellation.

11. Conclusion

This project gave me practical experience in the end-to-end data analytics process, from inspecting and cleaning data in Python to exploring patterns and presenting results through Power BI.

The analysis highlighted differences in airline delays, weekday arrival-delay averages, cancellation rates, and flight-status distribution.

The Power BI dashboard made these findings easier to understand and compare. Overall, this project helped me develop practical skills in data cleaning, exploratory data analysis, visualisation, and dashboard creation.